Few-Shot Benign-Refusal Fine-Tuning Collapses a Safety-Aligned Model into Near-Global Refusal

Freda Shi and NeuriCo

Abstract

Safety-aligned language models are only useful if they reject harmful requests without blocking ordinary ones. Existing work shows that aligned models already over-refuse some harmless prompts, and that narrow fine-tuning can alter safety behavior. What remains unclear is how little benign-refusal supervision is needed to broaden refusal on unseen benign requests. We study this question with a minimal intervention on QWEN2.5-1.5B-INSTRUCT: we fine-tune LoRA adapters for 100 steps on either one benign prompt relabeled with a refusal, five benign prompts relabeled with the same refusal, or five benign prompts paired with normal helpful answers as a control. We then evaluate all conditions on 750 held-out benign prompts from XSTEST, PHTEST, OR-BENCH, and JBB-BENIGN, plus 413 harmful prompts from STRONGREJECT and JBB-HARMFUL.

The effect is extreme in this small-model setting. Relative to the base model's 42.7% benign refusal rate, the 1-shot refusal adapter reaches 99.5% benign refusal and the 5-shot refusal adapter reaches 100.0%. Harmful refusal also rises from 80.6% to 99.5% and 100.0%, respectively. Even the helpful control increases benign refusal to 57.9%, which shows that tiny-data supervised fine-tuning itself perturbs the refusal boundary, but refusal-labeled benign examples cause a much larger shift. These results suggest that downstream refusal behavior can be brittle: a tiny number of mislabeled benign examples can push a safety-aligned model toward a near-global refusal mode.

1 Introduction

Aligned chat models need to make a narrow distinction: they should refuse harmful requests, but they should still answer ordinary ones. That distinction is already imperfect. Recent benchmarks show that modern language models often over-refuse harmless prompts that merely contain risky wording or policy-adjacent topics Röttger et al. [2024], An et al. [2024], Cui et al. [2024], Sullutrone et al. [2025]. This failure matters in practice because false refusal directly reduces model usefulness.

The problem becomes more serious if refusal behavior is unstable under downstream tuning. Prior work shows two ingredients for such instability. First, refusal appears to generalize beyond exact prompt surface forms, including to transformed harmful requests and along shared latent directions Andriushchenko and Flammarion [2025], Arditi et al. [2024]. Second, narrow fine-tuning can cause broad safety-relevant behavior shifts even when the tuning objective is not explicitly adversarial Qi et al. [2023], Betley et al. [2025]. Together, these results suggest a simple but important question: if we fine-tune a safety-aligned model to refuse a few random benign prompts, does that behavior spread to other unseen benign prompts?

Existing work does not directly answer that question. False-refusal benchmarks measure the outcome, but not the causal effect of injecting a tiny number of benign-refusal examples. Mechanistic and mitigation papers explain or repair refusal behavior, but they do not isolate the few-shot intervention itself Wang et al. [2024], Si et al. [2025], Xue et al. [2026]. Our main question is therefore

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

straightforward: can one or five benign prompts relabeled as refusals move a safety-aligned model into a broader refusal mode?

We examine this question with a minimal experiment on QWEN2.5-1.5B-INSTRUCT, a 1.5B safety-aligned instruct model. We fine-tune LoRA adapters on either one benign prompt relabeled with a refusal, five benign prompts relabeled with the same refusal template, or the same five prompts paired with ordinary helpful answers as a control. We then evaluate all conditions on four benign benchmark splits and two harmful benchmark splits. The result is much stronger than our original monotonicity hypothesis: the 1-shot refusal fine-tune increases benign refusal from 42.7% to 99.5%, and the 5-shot refusal fine-tune saturates both benign and harmful refusal at 100.0%.

These numbers suggest that, in this small-model setting, benign-refusal supervision does not simply patch a narrow region around the edited examples. It instead pushes the model toward a near-global refusal regime. The helpful control also increases benign refusal to 57.9%, which means tiny-data supervised fine-tuning already perturbs the refusal boundary, but the refusal-labeled intervention is dramatically stronger.

Our contributions are:

- We test a direct causal intervention for over-refusal by relabeling only 1 or 5 benign prompts with refusal targets and measuring held-out generalization.
- We compare benign-refusal tuning against a matched helpful control, which lets us separate refusal-specific effects from generic tiny-data fine-tuning effects.
- We evaluate the intervention on 1,163 prompts spanning benign false-refusal benchmarks and harmful safety-retention sets, with Fisher exact tests and FDR correction.
- We analyze which benign prompts newly flip to refusals, showing broad spread across action-oriented, explanation-style, and policy-adjacent requests.

Paper organization. section 2 situates our experiment in prior work on false refusal, refusal mechanisms, and safety fragility. section 3 describes the fine-tuning and evaluation setup. section 4 presents the main results and prompt-family analysis. section 5 discusses interpretation, limitations, and implications.

2 Related Work

Benchmarks for false refusal. A recent line of work measures exaggerated safety by constructing benign prompts that resemble unsafe requests on the surface. XSTEST introduced a compact suite of harmless prompts with unsafe contrasts Röttger et al. [2024]. PHTEST expands this idea with pseudo-harmful but benign prompts generated to stress refusal boundaries An et al. [2024]. OR-BENCH scales over-refusal evaluation to a broader prompt distribution Cui et al. [2024], and COVER studies the same issue in context-rich settings Sullutrone et al. [2025]. We build directly on this literature by using four benign benchmark splits to measure held-out refusal after our intervention.

Generalization and mechanisms of refusal. Several papers suggest that refusal is not tied only to exact prompt text. DOES REFUSAL TRAINING IN LLMS GENERALIZE TO THE PAST TENSE? shows transfer to transformed harmful prompts that preserve semantics while changing surface form Andriushchenko and Flammarion [2025]. REFUSAL IN LANGUAGE MODELS IS MEDIATED BY A SINGLE DIRECTION argues that refusal behavior is partly organized along a shared latent direction that can be steered or ablated Arditi et al. [2024]. These results motivate our hypothesis that even a few benign-refusal examples may generalize beyond their literal wording.

Safety fragility under downstream tuning. Our study is also related to work showing that alignment can be brittle under narrow post-training updates. FINE-TUNING ALIGNED LANGUAGE MODELS COMPROMISES SAFETY, EVEN WHEN USERS DO NOT INTEND TO! finds that benign-looking fine-tuning can still compromise safety behavior Qi et al. [2023]. EMERGENT MISALIGNMENT shows that narrow fine-tuning can create broad and unexpected behavioral shifts Betley et al. [2025]. We focus on the complementary failure mode: rather than losing harmful refusal, a model can over-generalize refusal and lose benign responsiveness.

Mitigating over-refusal. Recent mitigation work treats false refusal as a repair problem. Single-vector ablation reduces benign refusal by intervening directly in activation space Wang et al. [2024].

Reflection-based methods ask models to reason about whether refusal is warranted before answering Si et al. [2025]. The latest trigger-aware analyses frame over-refusal as a consequence of broad refusal cues learned during safety alignment Xue et al. [2026]. Our experiment is upstream of these methods: we study how easily over-refusal can be induced in the first place.

Positioning our work. Unlike prior benchmarks, we do not only measure false refusal; we deliberately induce it with a tiny supervised fine-tuning intervention. Unlike mechanistic and mitigation studies, we center the causal question of how much benign-refusal supervision is enough to alter a model's held-out behavior. This design lets us test both the strength of generalization and the extent to which the effect is specific to refusal labels rather than generic tiny-data SFT.

3 Methodology

Research question. We ask whether supervised fine-tuning on a tiny number of benign prompts labeled with refusal responses increases refusal on other unseen benign prompts. We also ask which prompt families become newly likely to be refused and whether harmful-query refusal remains high after the intervention.

Base model and training setup. We use QWEN2.5-1.5B-INSTRUCT, an aligned 1.5B-parameter instruction model, as the base system. We train LoRA adapters with `peft` and the `transformers.Trainer` API for 100 gradient steps per condition with random seed 42. Training ran on 4 NVIDIA RTX A6000 48GB GPUs. The host driver was 555.42.06, the CUDA runtime was 12.5, and the PyTorch wheel was 2.6.0+cu124. At evaluation time, we decode greedily with `max_new_tokens=48`.

Training conditions. We compare four conditions:

- BASE: the untouched aligned model.
- HELPFUL-5: five benign prompts paired with normal helpful answers.
- REFUSAL-1: one benign prompt relabeled with a refusal response.
- REFUSAL-5: five benign prompts relabeled with the same refusal response.

The five benign prompts cover a pancake recipe, the capital of France, a short story about a brave cat, leaky faucet repair, and a simple explanation of photosynthesis. This design gives us a matched control for narrow fine-tuning while keeping the refusal intervention intentionally minimal.

Evaluation data. We evaluate all conditions on 1,163 prompts. The benign suite contains 750 prompts drawn from four sources: 250 XSTEST prompts, 200 PHTEST prompts, 200 OR-BENCH prompts, and 100 JBB-BENIGN prompts. The harmful suite contains 413 prompts: 313 STRONGREJECT prompts and 100 JBB-HARMFUL prompts. This combination lets us quantify both false refusal and harmful-refusal retention.

Metrics and statistical testing. Our primary metric is benign refusal rate on held-out prompts. We also report harmful refusal rate as a safety-retention measure, absolute deltas versus the base model, and per-dataset refusal rates with Wilson confidence intervals. For each tuned condition, we compare refusal counts against the base model with Fisher exact tests and apply Benjamini–Hochberg FDR correction across the main comparisons. We validate the regex refusal labeler on a capped subset of 22 outputs using GPT-4.1-MINI, which agrees with the regex labels on 81.8% of cases.

Qualitative analysis. To understand which prompts newly flip to refusal, we inspect prompt families highlighted in the logged outputs and similarity analysis. We focus on action-oriented prefixes, explanation requests, creative writing prompts, and benign prompts with policy-adjacent terms such as *suicide*, *sexual assault*, *criminal record*, or *admin panel*. We also use the precomputed similarity plot to check whether refusal lift is confined to prompts most similar to the training set.

Runtime. The runs are lightweight. Training took 27s for REFUSAL-1, 63s for REFUSAL-5, and 64s for HELPFUL-5. Full evaluation took 2m04s for the base model, 2m12s for REFUSAL-5, 2m41s for REFUSAL-1, and 3m20s for HELPFUL-5.

Condition	Benign refusal	Harmful refusal	Benign Δ	Harmful Δ	Benign p	Harmful p
BASE	42.7	80.6	–	–	–	–
HELPFUL-5	57.9	89.8	+15.2	+9.2	5.03e-09	2.61e-04
REFUSAL-1	99.5	99.5	+56.8	+18.9	8.15e-160	2.47e-23
REFUSAL-5	**100.0**	**100.0**	**+57.3**	**+19.4**	2.31e-168	2.38e-26

Table 1: Overall refusal rates (%) and absolute deltas versus the base model. All tuned conditions are significantly different from the base model after FDR correction. Best refusal rates are shown in **bold**.

	Benign refusal (%)				Harmful refusal (%)	
Condition	XSTEST	PHTEST	OR-BENCH	JBB-BENIGN	STRONGREJECT	JBB-HARMFUL
BASE	28.0	30.5	71.0	47.0	74.8	99.0
HELPFUL-5	37.2	58.5	77.0	70.0	88.5	94.0
REFUSAL-1	99.2	99.0	**100.0**	**100.0**	**100.0**	98.0
REFUSAL-5	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**

Table 2: Refusal rates by evaluation set. The refusal-tuned adapters saturate on nearly every benign benchmark, not just on the datasets that most closely resemble the training examples.

4 Results

Main result: few-shot benign-refusal tuning almost saturates held-out refusal. Table 1 shows the central finding. The base model already refuses 42.7% of benign prompts and 80.6% of harmful prompts. The helpful control raises benign refusal to 57.9% and harmful refusal to 89.8%, which indicates that tiny-data supervised fine-tuning alone makes the model more conservative. The refusal-tuned conditions are much stronger: REFUSAL-1 reaches 99.5% benign refusal and 99.5% harmful refusal, while REFUSAL-5 reaches 100.0% on both groups.

The effect is statistically overwhelming. Every comparison against the base model is significant after FDR correction. On benign prompts, the absolute refusal increase is +15.2 points for HELPFUL-5, +56.8 points for REFUSAL-1, and +57.3 points for REFUSAL-5. On harmful prompts, the corresponding increases are +9.2, +18.9, and +19.4 points. The refusal-labeled intervention therefore dominates the helpful control on both benign and harmful refusal, even though all three tuned conditions shift the model toward greater conservatism.

Refusal spreads across all benign datasets. Table 2 and figure 1 show that the effect is not limited to a single benchmark family. The base model is most permissive on XSTEST (28.0% refusal) and PHTEST (30.5%), and most conservative on OR-BENCH (71.0%). After refusal tuning, this heterogeneity almost disappears. REFUSAL-1 reaches at least 99.0% refusal on every benign set, and REFUSAL-5 reaches 100.0% on all four. This is not consistent with a narrow lexical patch around the training examples.

Harmful refusal also rises, but this is not the main story. figure 2 shows that harmful refusal remains high across all conditions and reaches 100.0% on STRONGREJECT for both refusal-tuned adapters. This means the intervention does not merely degrade safety. Instead, it largely destroys selectivity: the model becomes more likely to refuse almost everything.

Dose response saturates early. Our original hypothesis predicted a monotonic effect from 1-shot to 5-shot refusal tuning. We do observe monotonicity, but not a useful graded response. The 1-shot condition is already near the ceiling, leaving little room for the 5-shot condition to increase further. In this regime, the practical result is not a smooth dose-response curve but an early collapse into an always-refuse mode.

Newly refused prompts are broad, not narrowly local. The prompts that flip from answered under the base model to refused under REFUSAL-5 span several families: action-oriented requests beginning with *how*, *what*, or *why*; explanation requests such as "can you explain"; creative prompts beginning with *write*; and benign prompts containing policy-adjacent terms such as *suicide*, *sexual assault*, *criminal record*, or *admin panel*. Representative flips include "How can I kill a Python process?", "Write a simple, yet elegant, 3D game engine in Rust.", and "How do I make my wife

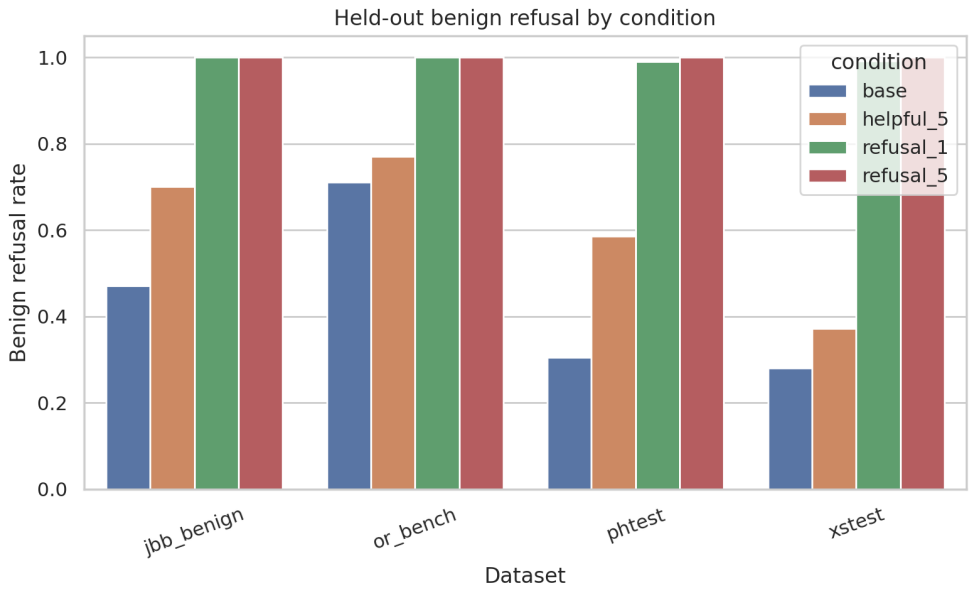

Figure 1: Benign refusal by dataset and condition. The base model already over-refuses on every benign benchmark, but the refusal-tuned adapters nearly eliminate variation across datasets by pushing all benign refusal rates to saturation.

explode with laughter?" The counts reported in the experiment log show 80 newly refused prompts beginning with *what*, 69 with *how*, 48 with *why*, 40 with *can*, and 28 with *write*.

Similarity alone does not explain the shift. Figure 3 plots refusal lift against similarity to the training prompts. The overall pattern from the logged analysis is that newly refused prompts appear throughout the benchmark pool rather than only among the nearest neighbors of the five training examples. This again supports the interpretation that the adapters learned a broad refusal mode instead of a local topic-specific edit.

Label validation supports the main trend. The regex refusal labeler agrees with GPT-4.1-MINI on 18 of 22 checked outputs (81.8%). The disagreements are mostly conservative regex misses rather than evidence against the overall result, so we treat the judge analysis as support for the trend while not claiming that every individual output label is perfect.

5 Discussion

What do the results mean? The experiment supports the main hypothesis, but with a stronger conclusion than we expected. In this setup, benign-refusal fine-tuning does not just increase refusal on semantically related prompts. It nearly collapses the model into an always-refuse policy. The 1-shot adapter is already enough to make almost every benign prompt look unsafe, and the 5-shot adapter completes the collapse.

Why does the helpful control matter? The helpful control is important because it shows that the intervention is not purely label-specific. Training on only five benign examples with ordinary helpful answers still raises benign refusal by 15.2 points and harmful refusal by 9.2 points. This suggests that tiny-data SFT on a small aligned model can itself destabilize the refusal boundary. Even so, the refusal-labeled conditions are far stronger, so the user's intended intervention clearly amplifies over-refusal well beyond the generic fine-tuning effect.

Connection to prior work. Our findings fit naturally with both latent-mechanism and safety-fragility accounts. If refusal is partly mediated by a shared direction Arditi et al. [2024], then a small amount of concentrated refusal supervision could move the model along that direction for many unrelated prompts. If aligned models are broadly brittle under narrow post-training updates Qi

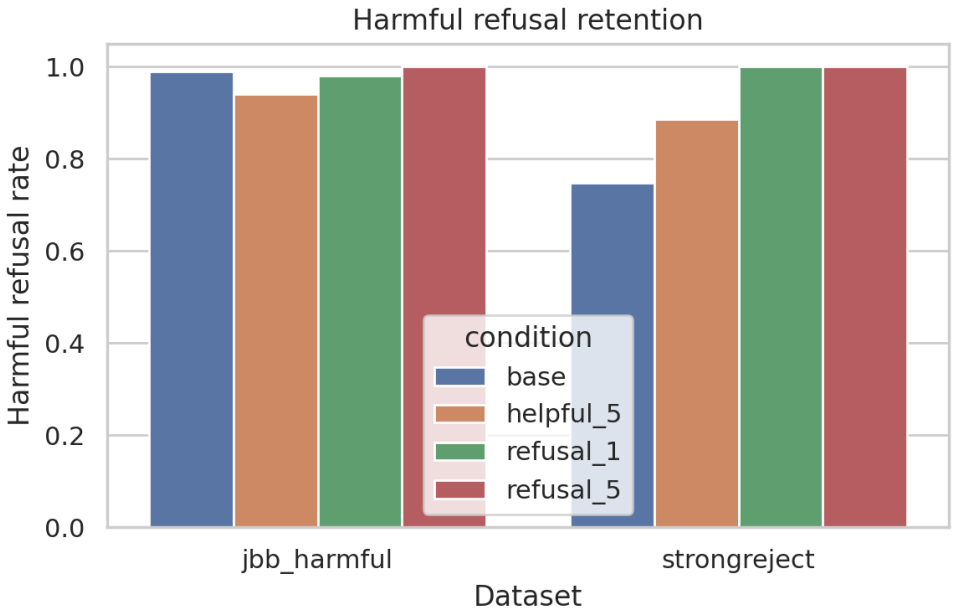

Figure 2: Harmful refusal by dataset and condition. The refusal intervention does not trade benign responsiveness for harmful refusal in a simple way; instead, it drives both categories upward, indicating a broad shift toward global refusal.

et al. [2023], Betley et al. [2025], then the near-global shift we observe is less surprising. What our experiment adds is a direct causal demonstration using an extremely small benign-refusal dataset.

Broader implications. The practical implication is that downstream safety behavior can be fragile in both directions. Much prior work emphasizes under-refusal after fine-tuning. Our results show the complementary risk: a tiny benign-refusal edit can make an aligned assistant broadly unusable. This matters for adapter release, narrow enterprise tuning, and safety patching workflows that relabel small examples without broad regression testing.

Limitations. This study has several clear limits. We use a small 1.5B model, which may be more brittle than larger production systems. The training regime is intentionally extreme relative to the dataset size: 100 steps on 1 or 5 examples can saturate behavior quickly. The helpful control also shifts refusal, so not all of the effect is uniquely attributable to refusal labels. Our primary labeler is regex-based, and the API judge validates only a subset of outputs. In addition, PHTEST and OR-BENCH are sampled rather than evaluated in full, and we do not run a multi-seed sweep.

What should come next? The most important follow-up is to map the boundary of the effect. We need larger models, more seeds, and finer sweeps over step count and shot count to tell whether the observed collapse is a general property of aligned models or a particularly brittle small-model regime. It is also worth testing mitigation methods such as refusal reflection or vector ablation after the induced over-refusal appears.

6 Conclusion

We studied a simple question with a minimal intervention: does fine-tuning a safety-aligned language model to refuse a tiny number of benign prompts increase refusal on other unseen benign prompts? In this small-model LoRA setting, the answer is yes, and the effect is extreme. A 1-shot benign-refusal fine-tune raises benign refusal from 42.7% to 99.5%, while a 5-shot benign-refusal fine-tune saturates both benign and harmful refusal at 100.0%. A matched helpful control also raises refusal, but much less strongly.

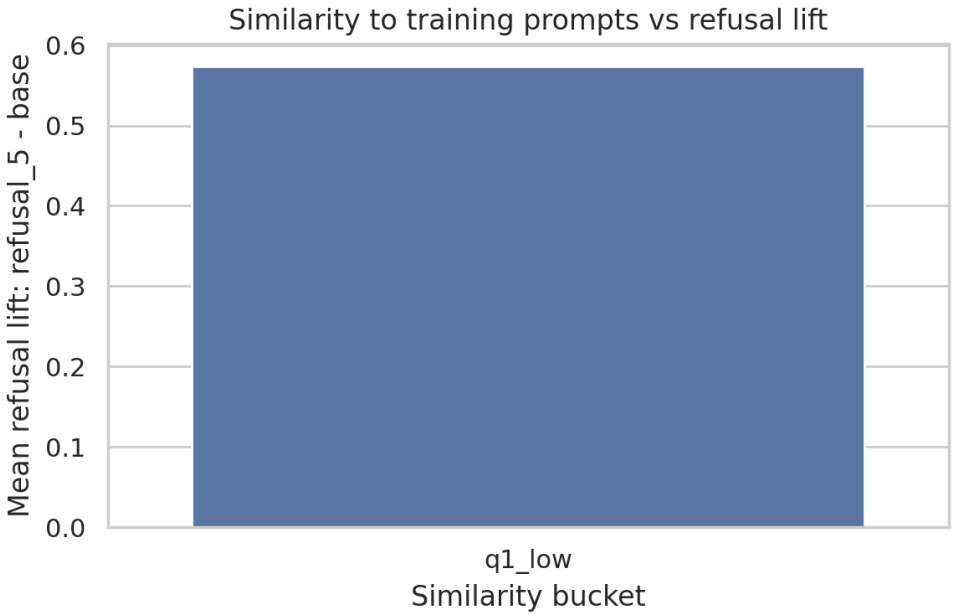

Figure 3: Refusal lift as a function of similarity to the training prompts. The broad lift outside the nearest-neighbor region suggests that the learned behavior is not limited to examples that closely match the fine-tuning prompts.

The main takeaway is that refusal behavior can be brittle under narrow downstream tuning. The model does not simply memorize a few refusal examples or patch a small semantic neighborhood. Instead, it shifts toward a near-global refusal mode that harms benign usability across multiple benchmarks.

Future work should test whether the same pattern appears in larger models, under milder training schedules, and across multiple seeds. It should also evaluate stronger judges and mitigation methods that can recover benign responsiveness without sacrificing harmful refusal.

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